

JAYAKISHAN B

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PROFESSIONAL SUMMARY

Pre-final year B.Tech Information Technology student passionate about AI Engineering, Generative AI, and Full-Stack Development. Experienced in developing AI-powered applications and machine learning solutions through internships, consultancy projects, and hackathons. Interested in building practical, user-focused AI systems and excited to learn, contribute, and grow in collaborative engineering environments.

EDUCATIONAL QUALIFICATION

Sri Ramakrishna Engineering College, Coimbatore - B.Tech Information Technology 2023 - 2027
CGPA: 8.44/10 (till 5th semester)

SBOA Matriculation and Higher Secondary School, Coimbatore - HSC 2022 - 2023
Percentage: 81.9%

SBOA Matriculation and Higher Secondary School, Coimbatore - SSLC 2020 - 2021
Percentage: 100%

EXPERIENCES

AI and Cloud Intern – Edunet Foundation and IBM SkillsBuild Sept 2025 - Oct 2025

- Completed an intensive 4-week virtual internship, finishing 8 structured AI certification modules covering Machine Learning, Deep Learning, NLP, Computer Vision, Prompt tuning, AI ethics and Model execution workflows.
- Built a capstone project (CareerForge – AI-powered ATS resume builder) integrating LLM-based resume optimization using prompt engineering and AI APIs with a full-stack application using Next.js and FastAPI.
- Implemented AI-driven resume analysis and recommendation workflows, generating personalized improvement suggestions and ATS optimization insights using LLM-powered evaluation pipelines.

Mobile App Development Intern – Resilience Business Grids LLP Dec 2024 - Jun 2025

- Built and deployed an AI-powered multimodal chatbot using Flutter and Dart, supporting text, voice, image, and video inputs; **served 200+ active users** and **handled 100+ concurrent queries**, reducing manual support workload by **30%** through automated intent classification and routing.
- Designed and implemented a scalable backend architecture with REST APIs, Firebase authentication, and data persistence; achieved **35% reduction in system latency** and **40% improvement in application performance**, enabling production-ready cross-platform deployment.

PROJECTS

AI-Powered ICU Patient Flow Coordinator *GE HealthCare - Consultancy Project*

- Designed and developed an AI-powered ICU decision support system using ensemble ML on **5,000+ patient records**, achieving **78% accuracy** and improving triage efficiency by **25–30%**.
- Developed a full-stack application using **Next.js, FastAPI, and SQLite** with REST APIs enabling concurrent access with real-time patient prioritization.
- Integrated explainable AI (SHAP) for transparent predictions, improving trust in clinical decision-making.

SREC Utsava 2026 - Symposium Platform *College Project*

- Engineered and deployed a scalable full-stack platform using **Next.js, Supabase, and Tailwind CSS** with role-based authentication and payment integration, serving **12,000+ users**.
- Optimized UI and backend APIs achieving **90+ Lighthouse scores** and handling **98,000+ page views** efficiently.
- Implemented role-based dashboards for admins and users enabling real-time system management and analytics.

AI-Driven Drug Discovery and Genomic Analysis Framework *InnoHack'26 VIT Vellore - Finalist*

- Built an AI prediction system using **Random Forest, XGBoost, and Gradient Boosting** with RAG-based LLM explanations, ranking in **top 21 out of 225 teams**.
- Developed a full-stack pipeline using **FastAPI and Next.js** for real-time predictions with SHAP-based interpretability.

Explainable AI Driven MLOps Framework for Fair Loan Advisory *Team Project*

- Designed an AI-powered advisory system improving user trust by **30–40%** using explainable AI and fairness monitoring.
- Built a production pipeline using **FastAPI, XGBoost, and Supabase** with automated retraining, drift detection, and RAG-based recommendations.

- Implemented fairness monitoring using Demographic Parity and Equal Opportunity metrics alongside automated model retraining and drift detection for responsible AI deployment.

AI-Driven Nocturnal Elephant Detection and Deterrence System

Hackathon Project - Winner

- Built a real-time AI-based detection and deterrence system using **YOLOv8n** and **OpenCV** achieving **85% accuracy** on edge devices.
- Integrated SMS/voice alerts using **Twilio API** with a monitoring dashboard for rapid response.

AI-Powered Cyber Triage System for Real-Time Threat Detection

Team Project

- Built a real-time threat detection system using **Random Forest** and **Flask** achieving **80% accuracy** and reducing response time by **40%**.
- Developed a network analysis pipeline with automated IP blocking and monitoring dashboard using **Wireshark**, **Scapy**, and **SQLite**.

PUBLICATIONS

- **Explainable AI Driven MLOps Framework for Fair and Inclusive Loan Advisory**
Accepted for presentation at the IEEE International Conference on Smart Communication and Sustainable Technologies (ICSCST 2026); publication forthcoming in IEEE Conference Proceedings.

HACKATHONS & AWARDS

- Finalist – InnoHack'26 (VIT Vellore) – Selected among top 21 teams from 225 nationwide entries in AI/ML track.
- Certificate of Recognition – GE HealthCare Consultancy – AI-powered ICU Patient Flow Coordinator.
- 1st Place – Human-Animal Conflict Hackathon – AI-Driven Nocturnal Elephant Detection and Deterrence System (Rs.25000 cash prize).
- Winner – Innovators Icon Award, Hack24 – AI-powered Lung Disease Prediction System.
- 1st Place – Protovision, Technovate 2025 – AI-Powered Cyber Triage System for Real-Time Threat Detection.

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, C, C++, SQL, TypeScript, JavaScript.
- **AI & Machine Learning:** Machine Learning, Deep Learning, Generative AI, Computer Vision, Explainable AI (SHAP), Machine Learning, Deep Learning, Generative AI, Computer Vision, Explainable AI (SHAP), Prompt Engineering, LLM Applications, Retrieval-Augmented Generation (RAG), MLOps.
- **Web & Backend Development:** FastAPI, Flask, Node.js, REST APIs, Next.js, React, Tailwind CSS.
- **Databases & Cloud:** Supabase (PostgreSQL), MySQL, SQLite.
- **DevOps & Tools:** Vercel, Docker, Git, GitHub, CI/CD, GitHub Actions, Linux.
- **Frameworks & Libraries:** TensorFlow, PyTorch, Scikit-learn, YOLOv8, OpenCV.
- **AI Tools & APIs:** OpenAI API, Gemini API, Hugging Face Transformers, LLM Integration, AI Workflow Automation.

CO-CURRICULAR ACTIVITIES

- Executive Member – Research and Innovation for Student Empowerment (RAISE), SREC.
- Executive Member – Computer Society of India (CSI) Student Branch, SREC.
- Executive Co-Chairperson – Road Safety Vertical, YI Yuva Club, SREC (2024-2025).
- Active Member – Phorartz Club (Photography & Arts), SREC.